Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect — Architecture Governance
Version: 1.2
Date: August 2026
Status: Accepted
Part of: ACRME Architecture Decision Records — one of four standalone, self-contained records.

> **About ADRs.** An Architecture Decision Record captures a single significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original. This record is self-contained: it can be read without any companion document. Evidence tags: `[Documented]`, `[Decided]`, `[Derived]`, `[Assumed]` (see Appendix).

ADR-001 — Region Selection

Status: Accepted
Date: August 2026
Deciders: Principal Cloud Architect, Platform Engineering, DR Owner
Related constraints: HC-1..HC-10 (hard constraints); VR-1..VR-11 (validation rules)

Context

Customers request capacity by supplying either a specific Azure region or an Azure geography (e.g. “US”, “Europe”, “Middle East”). The engine must derive three environment regions — **Prod**, **NonProd/CVAL**, and **DR** — that satisfy isolation, resilience, capacity, quota, and data-residency requirements.

Forces at play:

- Azure regions differ in SKU availability, zone count, capacity headroom, and quota posture. A naive “pick the nearest region” approach silently produces single-zone or capacity-starved placements.
- Some regions are **Restricted Capacity Regions** (sovereign clouds, limited-SKU regions) that must never be selected by automated placement.
- Certain geographies (notably the **Middle East**, with only UAE North and Saudi Arabia Central) do not have enough Standard regions to satisfy a three-region model in-geo, forcing a governed cross-geo DR path.
- Placement decisions must be **deterministic, auditable, and replayable** — the same inputs must always produce the same output, and every decision must be reconstructable for compliance.

Decision

Adopt a **staged, constraint-then-score placement pipeline** driven by environment-type-specific scoring formulas:

1. **Stage 1 — Eligibility pre-filtering.** Reduce all Azure regions to Standard Capacity Regions within the customer’s geography.
    - **HC-9 STANDARD_REGION_ONLY** excludes Restricted regions before scoring. `[Decided]`

- **HC-8 GEOGRAPHY_CONTAINMENT** confines the Prod anchor to the customer's chosen geography. `[Derived]`

2. **Stage 2 — Hard-constraint gate.** Each surviving region must pass **HC-1..HC-7 and HC-10** (region separation, capacity floor, quota floor, DR separation class, zone availability, DR coverage floor, DR floor integrity, cross-geo extension approval). Any failure excludes the region from scoring entirely — hard constraints are binary gates, never scored penalties. `[Decided]`

3. **Stage 3 — Environment-type-specific scoring.** Rank survivors using three formulas sharing default weights (α=0.30, β=0.20, γ=0.25, δ=0.15, ε=0.10) but with env-type-specific semantics for α (capacity signal) and δ (DR readiness): `[Decided]`

   - **Prod:** `Prod_region = argmax(PS_Prod(r))` over eligible Standard regions in-geo.
   - **NonProd/CVAL:** `argmax(PS_NonProd(r))` over survivors, excluding the Prod region.
   - **DR:** `argmax(PS_DR(r))` over survivors, excluding the Prod region (but **may** share with NonProd).

4. **Selection order is sequential — Prod → NonProd → DR** — not joint optimization. With 3–4 regions the greedy sequential result equals the joint-optimal result in nearly all practical cases, while remaining transparent and auditable. `[Decided]`

5. **Middle East special handling.** Prod and NonProd are chosen in-geo via `argmax(PS_Prod)` over `{UAE North, Saudi Arabia Central}`; DR is assigned to **Belgium Central** via the only approved Cross-Geo Extension paths (Saudi Arabia → Belgium Central, UAE North → Belgium Central). Belgium Central must itself pass HC-1..HC-10; if it fails, the placement is rejected with an ops alert — the engine never silently substitutes another region. `[Decided]`

6. **Determinism & audit.** All candidate sets, sub-scores, the winning score, and the active `PlacementPolicy` version are written to the `OperationRecord` for replay (VR-8, VR-9). Even the 3-region edge case (single eligible candidate) runs the full scoring path so the score is logged as a capacity-health signal. `[Decided]`

7. **Operating mode.** In Phase 1 the engine runs region selection in **recommendation/shadow mode** — it produces and logs the ranked recommendation but does not autonomously place. Autonomous placement is gated on POC validation of the scoring formulas.

Region Classification Model (normative) Every Azure region carries exactly one classification tier, stored in `PlacementPolicy` as config-as-code (versioned, auditable, replayable) — classification is a **governance decision, not a live capability query**. `[Documented]`

Tier	Eligibility	Regions
Standard Capacity Region	Eligible for dynamic selection, scoring, and all env assignments (Prod/CVAL/DR) — the only regions that enter the pipeline	NA: West US 3, Central US, Canada Central · EU: Sweden Central, Belgium Central · ME: Saudi Arabia, UAE North · APAC: Japan East, Southeast Asia, Australia East
Restricted Capacity Region	Exception-only (Scenario 2 + Prod + approval); never scored, ranked, or recommended	East US 2, North Europe, West Europe, East Asia, Australia Southeast (all: Azure physical capacity constraint)

Tier	Eligibility	Regions
Cross-Geo Extension Region	DR-only, for geographies that cannot meet the 3-region minimum in-geo	Saudi Arabia → Belgium Central; UAE North → Belgium Central (Middle East only)

Restricted regions are excluded by a **pre-filter ahead of all hard constraints** (HC-9), never as a scoring penalty.

Prod Region Input Modes The Prod region is the anchor; CVAL and DR are selected sequentially from it. The customer supplies the anchor one of two ways, and both converge on a single validated Prod anchor: `[Documented]`

- **Scenario 1 — geography supplied.** The engine derives Prod via `argmax(PS_Prod)` over Standard Capacity Regions **in that geography only**. Ties break deterministically by the Standard-region list order for the geography; the first-listed region is the deterministic cold-start default when no snapshot exists. Geography exhaustion → reject (never silently cross-geo or use a Restricted region).
- **Scenario 2 — specific region supplied.** If Standard, validate against HC-1..HC-10 and adopt as the anchor (`PS_Prod` used only for post-selection validation). If Restricted, route to the Exception Deployment Workflow.

Exception Deployment Workflow (Scenario 2 — Restricted region) A Restricted region is used only if **all four** conditions hold; failure rejects at the first failing gate: `[Documented]`

#	Condition	Check
EC-1	Explicit request	Customer named the region; engine never recommended it
EC-2	Production only	CVAL and DR may **never** use a Restricted region
EC-3	Exception approval	A named, revocable approval record exists for the customer–region pair
EC-4	Scenario 2 input	Restricted regions can never be engine-derived

On success the region becomes the **Exception Prod Anchor**, the placement is marked an **Exception Deployment**, a capacity-constraint warning is emitted to the caller, and the approval ID + restriction status are mandatory `OperationRecord` fields (commit blocked if absent). CVAL/DR still select from Standard regions via normal scoring.

Validation Rule Framework (VR-1..VR-11)

Rule	Check	Failure action
VR-1	Region exists in classification list	Reject if unknown
VR-2	Automated placement uses Standard only	Exclude before scoring if Restricted

Rule	Check	Failure action
VR-3	Scenario 2 Restricted: EC-1..EC-4 all met	Reject at first failing condition
VR-4	Scenario 1 derived Prod in-geo Standard	Geography-scoped exhaustion error
VR-5	Standard region passes HC-1..HC-10	Exclude; exhaustion error if all excluded
VR-6	Middle East DR = Belgium Central via approved path	Block with ops alert if Belgium Central fails HC-1..HC-10
VR-7	Exception approval ID persisted before commit	Block commit if absent
VR-8	Capacity-constraint warning emitted	Block commit if suppressed
VR-9	Snapshot age within policy limit	Trigger targeted ARM refresh
VR-10	Capacity hold acquired before commit	Block commit if hold absent
VR-11	Restricted regions absent from all recommendation outputs	Post-scoring filter (defence-in-depth)

Capacity Holds & Concurrency Before returning a committed assignment the engine creates a **capacity hold** keyed by region, SKU, zone, environment, and policy version, using **optimistic concurrency**; the hold expires if Azure provisioning does not begin. This closes the concurrent-placement race. `[Documented]`

Corrected Scoring Model (pilot)

- Default weights retained for pilot comparison: `α=0.30, β=0.20, γ=0.25, δ=0.15, ε=0.10`; every component is clamped `Clamp(x) = max(0, min(1, x))`. `[Assumed]`
- To avoid double-counting the same signal under α and δ, the CVAL/NonProd formula is proposed as `PS_NonProd = 0.35·Capacity + 0.25·Quota + 0.25·Distribution + 0.05·DR_Overflow_Integrity + 0.10·Zones`. `[Assumed]`
- Distribution uses **demand units, not customer count**: `Distribution = 1 − Region_Assigned_Demand / Total_Assigned_Demand`. `[Assumed]`
- Revised weights are proposed, not empirically validated — advisory until pilot measurement.

Governance & Compliance Controls

- The classification list lives in `PlacementPolicy`; any change requires a policy-version increment, a Decision Log entry, and **replay of the prior 30 days of placements** against the new classification before activation. `[Documented]`
- Exception approval records are revocable engine artefacts; a revoked approval blocks future exception deployments for the customer–region pair with no code change.
- Every classification change is audited with approver identity, timestamp, previous/new classification, and affected geography.
- Belgium Central's regional capacity-planning targets must include potential Middle East DR demand on top of in-geo Europe demand.

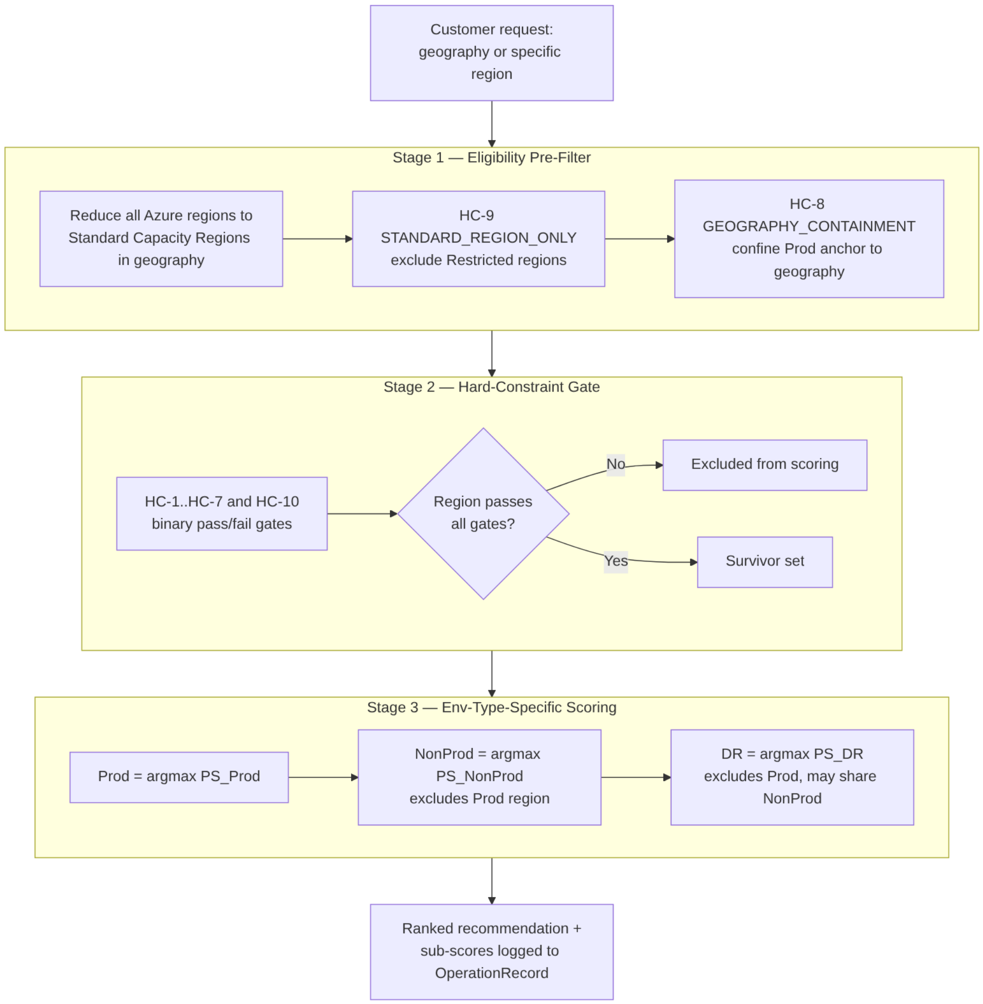

Figure 1: **Figure 1.** ACRME staged placement pipeline — Stage 1 eligibility pre-filter, Stage 2 hard-constraint gate, Stage 3 environment-type-specific scoring, then sequential Prod → NonProd → DR selection.

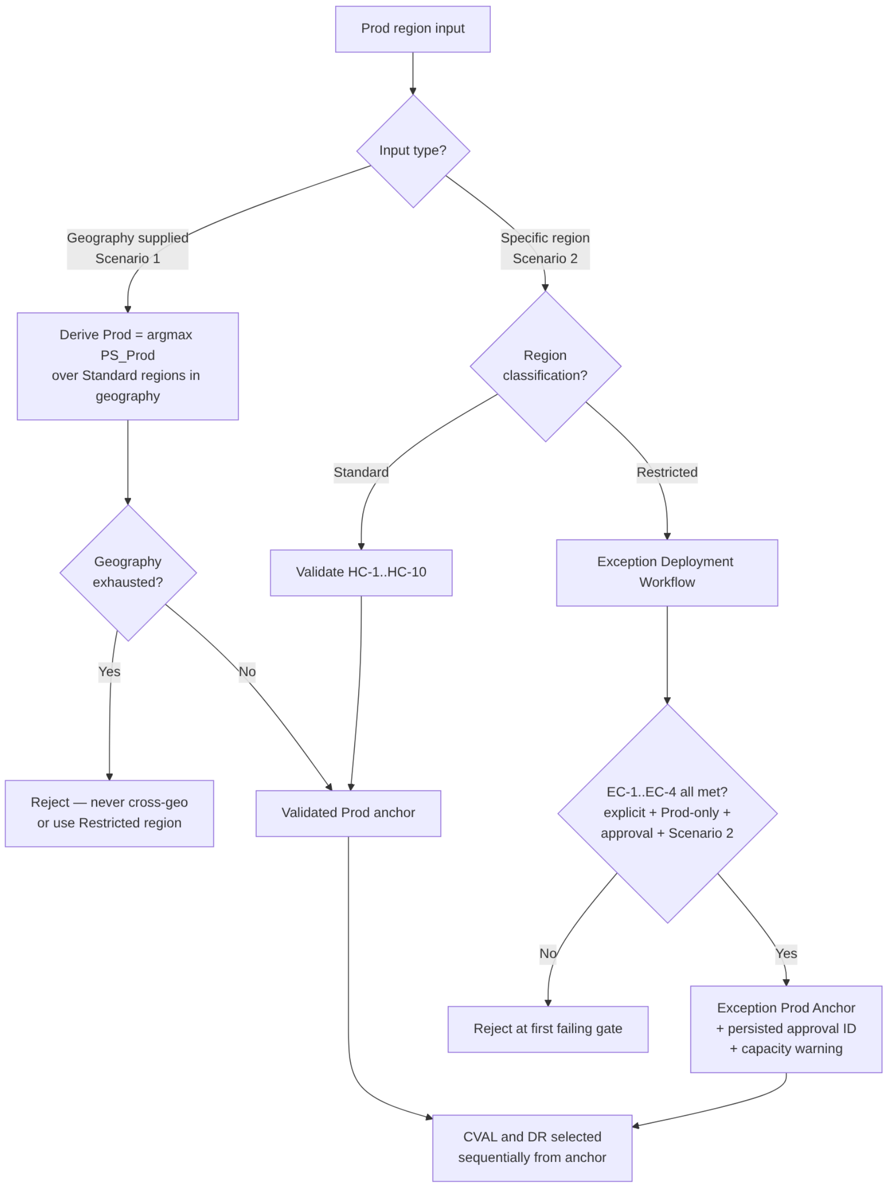

Figure 2: **Figure 2.** Prod region input modes — geography-supplied (Scenario 1) versus specific-region (Scenario 2), including the restricted-region Exception Deployment Workflow.

Consequences

Positive: - Deterministic, replayable placement with a complete audit trail satisfies compliance and post-incident review. - Hard constraints guarantee no placement ever violates isolation, zone, quota, or residency rules regardless of score. - Restricted regions are structurally impossible to select automatically; exceptions flow through a governed Scenario 2 workflow with a persisted, revocable approval ID. - Env-type-specific formulas let Prod optimise for capacity/isolation while DR optimises for overflow readiness.

Negative / trade-offs: - Three formulas plus 16 per-CRG-type `RegionalSnapshot` fields increase implementation and snapshot-maintenance cost. - Sequential selection is greedy; if the region count ever exceeds 6, joint optimization should be revisited. - Cross-geo extension paths are a manual governance artefact — adding a path requires a PlacementPolicy update, governance approval, and a Decision Log entry. - Worked scoring examples remain a known gap until per-CRG-type inputs are finalised.

Neutral: - CRG_Score (Regional Capacity Weight) is demoted from a primary scoring input to a monitoring-only signal.

Alternatives Considered

Alternative	Why Rejected
Joint optimization (select all three regions simultaneously)	Higher combinatorial complexity; opaque and harder to audit; identical result to sequential for 3–4 regions. `[Decided]`
Single generic PS(r, env_type) formula	α and δ have fundamentally different meanings per env type; a single formula needs so many branches it becomes three formulas anyway. `[Decided]`
Geographic distance as a scored soft objective	Operators already choose geographically independent region sets at design time; HC-4 as a hard constraint is sufficient and simpler. `[Decided]`
Silent fallback region for failed Middle East DR	Violates governance and residency guarantees; the engine must fail loudly and require an approved extension path. `[Documented]`

Appendix — ADR Summary

ADR	Hard Constraints Applied	Key Open Items
ADR-001 Region Selection	HC-1, HC-4, HC-5, HC-8, HC-9, HC-10	Worked scoring examples pending final per-CRG-type inputs

Appendix — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR

Appendix — Evidence Tag Taxonomy

Tag	Meaning
`[Documented]`	Traceable to Azure platform behaviour or documentation
`[Decided]`	An explicit ACRME design choice recorded in this ADR set
`[Derived]`	A logical consequence of a documented constraint or decision
`[Assumed]`	Architectural judgement pending proof-of-concept validation

Related ADRs

- **ADR-002 — Quota and Capacity Management** (`acrme_adr_002_quota_and_capacity_management.md`
- **ADR-003 — Capacity Management during Disaster Recovery (DR)** (`acrme_adr_003_capacity_mana`
- **ADR-004 — Forecast and Increase of Capacity and Quota** (`acrme_adr_004_forecast_and_increase_`

Document Status: Accepted
Next Review: After proof-of-concept validation of Azure Quota Groups GA and quota-release latency, and on resolution of the consumer-credential model and engine-mode state-machine items.